ANDREA SOLFRIZZI

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ABOUT ME

Master's student in Advanced Automotive Engineering, with a deep passion for **motorsport**, **simulation modeling**, and a distinct focus on **aerodynamic** studies. My knowledge in fluid dynamics has been shaped by specialized courseworks at the **Dallara Academy** and an advanced **Turbulence** course during my master's studies, which effectively complement my academic foundation in **CFD and Aerodynamics**.
Furthermore, my active role in the **Formula Student** program's Dynamics division has allowed me to hone my skills in vehicle dynamics and team collaboration.
Highly motivated and focused on problem-solving, I am eager to apply my technical background to real-world challenges in the high-performance automotive sector.
Productive, organized and motivated toward learning and continuous improvement.

EDUCATION

Master's Degree in Advanced Automotive Engineering - High Performance Car Design (LM-33)

MotorValley University of Emilia-Romagna

September 2024 – Today
Modena/Varano de' Melegari, Italy

Main topics:
- Aerodynamics of High Performance and Racing Cars at Dallara Academy
- CFD process, WT workshops at Dallara Academy
- Turbulence: Kolmogorov Theory, Wall Turbulence, Free-shear flows, Forced and Natural Convection
- Vehicle Dynamics
- FEM and Chassis Fundamentals
- CAD and Design of Product
- NVH testing
- Manufacturing and Assembly Technologies

Bachelor's Degree in Automotive Engineering (L-9)

University of Modena and Reggio Emilia

September 2020 – July 2024
Modena, Italy

Score: **110/110 cum laude**

Main topics:
- Fluid Dynamics and Thermodynamics
- Theory of Structures
- Mathematical Methods for Engineering
- Engineering Materials
- Engines and Machines

CERTIFICATIONS

Cambridge B2 First
06/2018 | **Score: 182/190**

LANGUAGES

Italian (Native)	●●●●●
English	●●●●◐
Spanish	●●◐○○

HARD SKILLS

3DEXPERIENCE®Platform: CATIA v5/SolidWorks
Design of mechanical tolerancing

Hexagon/MSC®ADAMS
Parallel, Opposite Wheel travel, track simulations, hardpoints modifications

OpenModelica®
Lumped Parameter modelling through the SFPLibDyn library

MathWorks®MATLAB
Controllers, Dynamic Modelling, coding

C++ and Python
Basic understanding of main libraries and coding

Hexagon/MSC®Marc Mentat
Running and setting a FEM simulation, interpreting data and model modification, boundary condition setting

PROJECTS

Formula Student: Vehicle Dynamics division

Determination of a vehicle dynamic model of the Formula SAE MMR Hybrid car, with identification of the ground clearances and stiffnesses in varying driving conditions (straight running, braking, acceleration), and in varying setup configurations.

- **Tools:** MS Excel

Brake Squeal: Literature Review And Methods of Analysis

Authored a comprehensive analytical review of automotive brake squeal, evaluating the theoretical mechanisms, experimental testing procedures, and numerical approaches used to study and predict friction-induced vibrations. The research focused on understanding the mode-coupling instability phenomenon by assessing various mathematical lumped-parameter models, ranging from simple 2-DOF to complex 5-DOF systems. Through this analysis, the project determined how factors such as non-linear contact stiffness, friction variability, and targeted damping strategies influence system resonance and overall acoustic stability.

Lumped Parameter Model of Porsche 911 GT3RS (992) DRS and Active Front Flaps

Developed a lumped parameter model of the hydraulic active aerodynamics for the Porsche 911 GT3 RS, focusing on the rear Drag Reduction System (DRS) and the theoretical conversion of the active front flaps to hydraulic actuation. The project involved designing innovative hydraulic control architectures—including dual servo-valve setups to compensate for differential piston areas and a load-sensing circuit to minimize energy losses—to maintain optimal aerodynamic balance. The system's viability for high-performance track applications was validated by analyzing flow distribution, pressure dynamics, and actuator response times.

- **Tools:** OpenModelica, MATLAB, MSC ADAMS, MS Excel

Decoupled Suspensions for a Formula SAE car

Engineered a fully decoupled rear suspension system for a Formula SAE vehicle to independently optimize heave and roll kinematics while ensuring regulatory compliance. The design features an unidirectional lever mechanism that maintains the roll spring in compression, alongside topology-optimized titanium rockers that achieved an 85% mass reduction. The assembly's structural integrity and high-performance reliability were rigorously validated through comprehensive stress, kinematic, and tolerance stack-up analyses.

- **Tools:** 3DEXPERIENCE®platform: CATIA, CAE softwares, MSC ADAMS, MS Excel

Wind Tunnel Workshop at Dallara

Collaborated with Dallara engineers in a specialized wind tunnel session to optimize the aerodynamic package of a Formula 3 2017 model. The initiative focused on evaluating four distinct aerodynamic configurations to maximize downforce while executing practical aero rebalancing and drag adjustment (redrag) techniques. This hands-on testing was underpinned by a dedicated Computational Fluid Dynamics (CFD) workshop, which involved analyzing simulation data to guide and validate the physical testing strategy.

HOBBIES AND INTERESTS

Working Out | Motorsports | Music | Movies